1 I

1.1 (1)

The Lagrangian is:

$$\mathcal{L} = T - U = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{\alpha}{r}.$$

Applying the Euler-Lagrange equation to r, we obtain:

$$m\ddot{r} = mr\dot{\theta}^2 + \frac{\alpha}{r^2}.$$

Similarly, applying it to θ:

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0.$$

1.2 (2)

Let $u = 1/r$. From the conservation of angular momentum:

$$L = mr^2\dot{\theta},$$

we have:

$$\dot{\theta} = \frac{L}{m}u^2.$$

Differentiating $r = 1/u$ with respect to time:

$$\dot{r} = -\frac{1}{u^2}\frac{du}{d\theta}\dot{\theta} = -\frac{L}{m}\frac{du}{d\theta}.$$

Differentiating once more:

$$\ddot{r} = -\frac{L}{m}\frac{d^2u}{d\theta^2}\dot{\theta} = -\frac{L^2}{m^2}u^2\frac{d^2u}{d\theta^2}.$$

Substituting this into the radial equation of motion:

$$-\frac{L^2}{m^2}u^2\frac{d^2u}{d\theta^2} = \frac{L^2}{m^2}u^3 + \frac{\alpha}{m}u^2.$$

Rearranging, we obtain:

$$\frac{d^2u}{d\theta^2} + u = -\frac{m\alpha}{L^2}.$$

The homogeneous solution is:

$$u_0 = A\cos(\theta - \theta_0),$$

and a particular solution is:

$$u^* = -\frac{m\alpha}{L^2}.$$

Thus, the general solution is:

$$u = u_0 + u^* = A\cos(\theta - \theta_0) - \frac{m\alpha}{L^2}.$$

Since $u = 1/r$, the orbit is given by:

$$\boxed{r = \frac{1}{A\cos(\theta - \theta_0) - \frac{m\alpha}{L^2}}}.$$

1.3 (3)

As $t \to -\infty$, the velocity is parallel to the incoming asymptote, and the perpendicular distance from the origin to this asymptote is b.

Therefore, the magnitude of the conserved angular momentum is:

$$\boxed{L = mv_0 b}.$$

1.4 (4)

For now, we choose the coordinate system such that $\theta_0 = 0$, which gives:

$$r = \frac{1}{A\cos\theta - \frac{m\alpha}{L^2}}.$$

As $t \to \pm\infty$, we have $r \to \infty$. Therefore, the denominator must vanish at the corresponding asymptotic angles $\theta = \pm\theta_1$:

$$A\cos\theta_1 - \frac{m\alpha}{L^2} = 0.$$

Hence:

$$\boxed{\cos\theta_1 = \frac{m\alpha}{AL^2}}.$$

1.5 (5)

Differentiating r with respect to time, we obtain:

$$\dot{r} = \frac{A\dot{\theta}\sin\theta}{\left(A\cos\theta - \frac{m\alpha}{L^2}\right)^2} = \frac{AL}{m}\sin\theta.$$

As $t \to -\infty$, the tangential velocity vanishes:

$$r\dot{\theta} = \frac{L}{mr} \to 0.$$

Hence, only the radial component of the velocity remains. Since the particle approaches the origin from the incoming asymptote, we have $\theta \to -\theta_1$ and $\dot{r} \to -v_0$.

Therefore:

$$-v_0 = -\frac{AL}{m}\sin\theta_1.$$

Thus:

$$\boxed{\sin\theta_1 = \frac{mv_0}{AL}}.$$

1.6 (6)

From the geometric relation:

$$\phi_s + 2\theta_1 = \pi,$$

we obtain:

$$\tan\frac{\phi_s}{2} = \cot\theta_1 = \frac{\cos\theta_1}{\sin\theta_1}.$$

Substituting the results from (4) and (5):

$$\tan\frac{\phi_s}{2} = \frac{\frac{m\alpha}{AL^2}}{\frac{mv_0}{AL}} = \frac{\alpha}{Lv_0}.$$

Finally, using $L = mv_0 b$:

$$\boxed{\tan\frac{\phi_s}{2} = \frac{\alpha}{mv_0^2 b}}.$$

2 II

2.1 (7)

The Lagrangian is:

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{\beta}{r^n}.$$

Applying the Euler-Lagrange equation to r:

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{n\beta}{r^{n+1}}.$$

Similarly, applying it to θ:

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0.$$

2.2 (8)

For uniform circular motion, we have:

$$r = r_0, \qquad \dot{\theta} = \omega_0, \qquad \dot{r} = \ddot{r} = 0.$$

Substituting these conditions into the radial equation:

$$mr_0\omega_0^2 - \frac{n\beta}{r_0^{n+1}} = 0.$$

Therefore:

$$\boxed{mr_0^{n+2}\omega_0^2 = n\beta}.$$

Since the angular momentum is conserved:

$$L = mr_0^2\omega_0.$$

We can also rewrite the above relation as:

$$\boxed{Lr_0^n\omega_0 = n\beta}.$$

2.3 (9)

Since the angular momentum remains unchanged after the perturbation, we have:

$$L = mr^2\dot{\theta} = mr_0^2\omega_0.$$

The radial equation can therefore be rewritten as:

$$m\ddot{r} = \frac{L^2}{mr^3} - \frac{n\beta}{r^{n+1}}.$$

Substituting $r = r_0 + \rho$, with $\rho/r_0 \ll 1$, we obtain:

$$m\ddot{\rho} = \frac{mr_0^4\omega_0^2}{(r_0+\rho)^3} - \frac{n\beta}{(r_0+\rho)^{n+1}}.$$

Expanding both terms to the first order in ρ/r_0:

$$\begin{aligned} m\ddot{\rho} = &\frac{mr_0^4\omega_0^2}{r_0^3}\left(1 - 3\frac{\rho}{r_0}\right) \\ &- \frac{n\beta}{r_0^{n+1}}\left(1 - (n+1)\frac{\rho}{r_0}\right). \end{aligned}$$

Using the circular orbit condition:

$$\frac{mr_0^4\omega_0^2}{r_0^3} = \frac{n\beta}{r_0^{n+1}} = mr_0\omega_0^2,$$

the constant terms cancel, leaving:

$$m\ddot{\rho} = mr_0\omega_0^2\left[-3\frac{\rho}{r_0} + (n+1)\frac{\rho}{r_0}\right].$$

Thus:

$$m\ddot{\rho} = (n-2)m\omega_0^2\rho.$$

Rearranging:

$$\boxed{\ddot{\rho} + (2-n)\omega_0^2\rho = 0}.$$

For the radial motion to be oscillatory, the coefficient of ρ must be positive. Therefore:

$$\boxed{0 < n < 2}.$$